

Chang Liu

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EDUCATION

University of Wisconsin-Madison

Madison, WI

Bachelor of Science, Computer Science

May 2026 (Expect)

Bachelor of Science, Data Science

May 2026 (Expect)

Related courses: Artificial Intelligence, Algorithm, Discrete Mathematics, Computer Engineering, Data Science Modeling, Big Data System, Human-Computer Interaction, Software Engineering, Machine Organization, User Experience Research.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, Python, Kotlin, SQL, R, C, C#, C++, HTML

Tools/Frameworks: Spring Boot, React, Mockito, PyTorch, Node.js base, RESTful, Git, GitHub, Figma, Jira, Android Studio, Cursor, Copilot, Docker, Kubernetes, Elasticsearch, Node.js, Linux, GCP, ROS, URDF

Languages: English, Chinese, Korean (Beginner)

PROFESSIONAL EXPERIENCE

ZhengGong Technology Co.Ltd. | Beijing, China

May 2024 – Aug 2024

System Development Intern

- I optimized the program logic by introducing a lightweight **Unsupervised Machine Learning** approach. Implemented an auto-tuning algorithm **using K-means clustering** to automatically adjust the sensitivity, so the system could adapt to different fluid conditions without manual calibration.
- Edge case testing** and supported the launch of a product line resulting approximately **\$25000 revenue**.

RedBing LLC | Atlanta, GA, USA

May 2025 – Aug 2025

Software Development Intern

- Assisted in develop and optimize a real-time search infrastructure using **Kafka**, **Elasticsearch**, **SQL** and **Spring Boot**, improving indexing latency and query performance.
- Implemented data processing pipelines using **Python** and **Flask** for cleaning, transformation, and streaming.
- Collaborated cross-functionally to **integrate backend search systems with frontend applications**, improving user-facing search experience.

RESEARCH EXPERIENCE

People and Robots Lab | University of Wisconsin-Madison | Madison, WI

Oct 2025 – Present

Undergraduate Researcher

- Built a **VR-based teleoperation system** integrating **Meta Quest 3** with a **UR3e robotic arm** using **ROS**, **MoveIt**, and **Unity**, enabling real-time control of physical hardware.
- Designed and debugged **TCP/IP communication** between **Linux** workstations and robotic systems, improving system stability under variable network conditions.
- Implemented **coordinate frame transformations** and **motion planning** pipelines, resolving inverse kinematics failures and unreachable pose errors.
- Integrated on-device **computer vision** (YOLO + Unity Sentis) for object detection to support assistive robotic tasks.
- Conducted user testing and iterated system design to improve usability and reliability in human-robot interaction workflows.

ACADEMIC PROJECT

Kafka Weather Data Streaming System | UW-Madison | Madison, WI

Jan 2024 – May 2024

- Designed a distributed real-time data pipeline using **Kafka** to simulate and process high-throughput weather data streams.
- Built **producer-consumer architecture** handling continuous data ingestion and transformation with low-latency processing.
- Implemented **fault-tolerant streaming** workflows and **JSON-based** reporting for downstream visualization systems.
- Containerized services using **Docker** to ensure reproducibility and scalable deployment across environments.

SeeSafe Mobile Application | UW-Madison | Madison, WI

Sep 2024 – Dec 2024

- As **Project Manager** and **backend developer**, organizing tasks and presenting team progress reports.
- Developed an **Android mobile application** using **Kotlin**, aimed at assisting individuals with vision impairments by connecting them with volunteers for real-time assistance and guidance.
- Implemented advanced GUI components, including dialogs, menus, and fragments, to enhance user accessibility.
- Integrated multiple **APIs** to implement features such as **real-time video calls**, push **notifications**, and Text-to-Speech functionality.

LEADERSHIP & INVOLVEMENT EXPERIENCE

Women Basketball Club | UW-Madison | Madison, WI

Sep 2022 – Present

- Co-founder & Vice President

Adventure Learning Program (ALPs) | UW-Madison | Madison, WI

Oct 2024 – Present

- Facilitator, Risk Management Committee